4.6 – Similarity and Transformations • Homework

Name: _____

Find the image after a dilation centered at the origin with the given scale factor.

1. $(-2, 3)$, $k = 5$

2. $(12, -8)$, $k = 1/4$

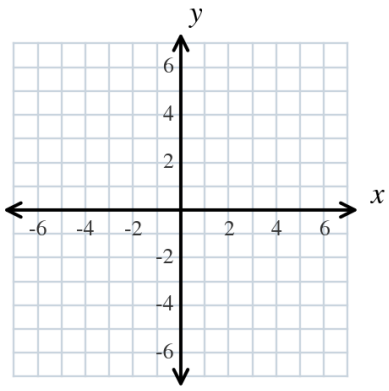
Name the similarity transformation that maps the preimage onto the image.

3. $\triangle ABC$ with $A(1, 1)$, $B(2, 4)$, $C(4, 2) \rightarrow \triangle A'B'C'$ with $A'(3, 3)$, $B'(6, 12)$, $C'(12, 6)$

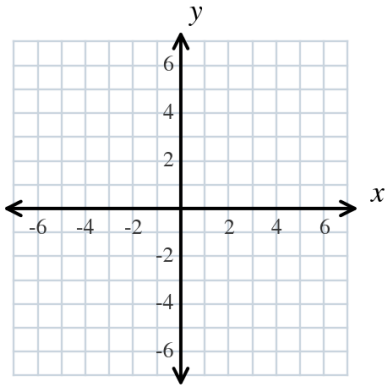
4. $\overline{DE}$ with $D(4, 6)$, $E(8, 2) \rightarrow \overline{D'E'}$ with $D'(2, 3)$, $E'(4, 1)$

Find the final image after each composition.

5. (3, 1) after the translation $(x, y) \rightarrow (x - 2, y + 4)$, then a dilation about the origin with $k = 2$.

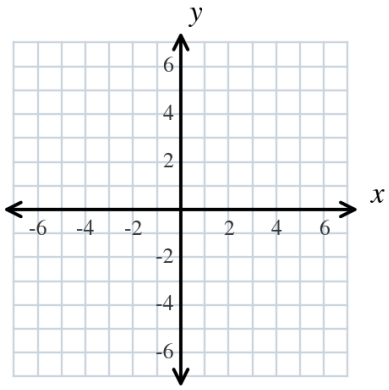


6. $(8, -12)$ after a dilation about the origin with $k = 1/4$, then a reflection in the x-axis.



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7. Graph △JKL with J(1, 1), K(2, 3), and L(4, 1) and its image after a dilation about the origin with k = 2, then a reflection in the y-axis. List the final coordinates.



Answer each question.

8. △MNP ~ △XYZ with a scale factor of 4. If MN = 2.5 and m∠N = 65°, find XY and m∠Y.

9. Two figures are similar with k = 1. What else can you say about them?

10. A student says every pair of similar figures must be the same size somewhere, because angles match. Explain the mistake.